Deep Neural Networks - Programming Assignment 2

Student and configuration

Souvik Dutta | 2026AJ05991

Problem 1 h=64; Problem 2 d=128, 4 heads; seeds 42, 43, 44. The withdrawn variants table was not used. Individual ablation: vanilla-RNN gradient clipping at T=200, norm 1.0 versus none.

Methods

UCI HAR uses nine channels, 128 steps and six activities with a subject-wise split and training-only normalization. The adding task uses two markers and the target sum. Scratch recurrent cells and Transformer attention/encoder blocks were tested against framework/reference calculations.

Execution evidence

Full experiments ran on a Colab Tesla T4 with PyTorch 2.14.1+cu130. CSVs, figures, samples and notebooks are included with the submission. Local regression suite: 7 passed.

Problem 1 - HAR classification and parameter accounting

Parameter counts

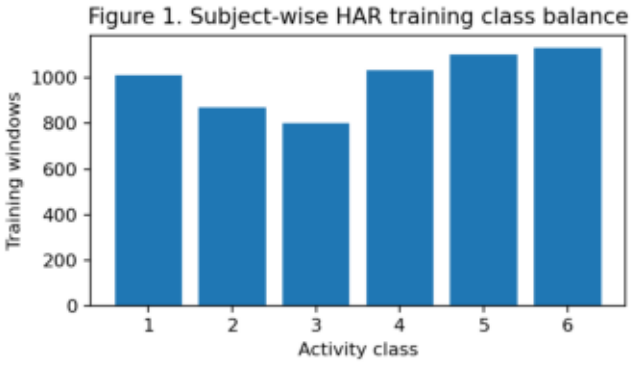
For input size 9 and h=64, one-bias equation counts are RNN 4,736, LSTM 18,944 and GRU 14,208. PyTorch stores two bias vectors: 4,800, 19,200 and 14,400.

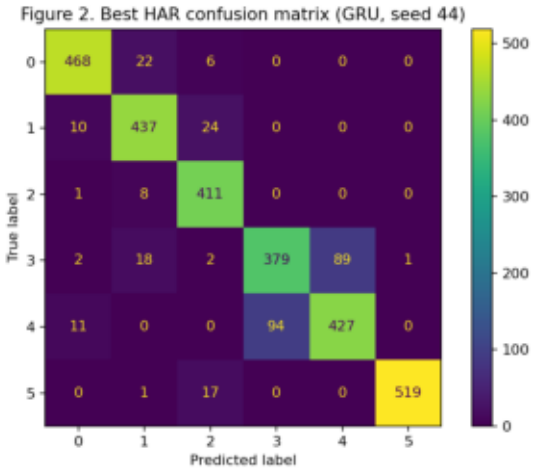
HAR results

RNN: accuracy 65.739 +/- 3.335%, macro-F1 64.994 +/- 3.338%, 5,190 parameters, 0.186 s/epoch. LSTM: 85.997 +/- 2.952%, 85.664 +/- 3.108%, 19,590 parameters, 0.183 s/epoch. GRU: 88.847 +/- 0.674%, 88.838 +/- 0.731%, 14,790 parameters, 0.176 s/epoch.

Interpretation

GRU is the strongest wearable choice: best accuracy and macro-F1, lower parameter count than LSTM, and lowest measured epoch time.





Problem 1 - Adding task

Results over sequence length

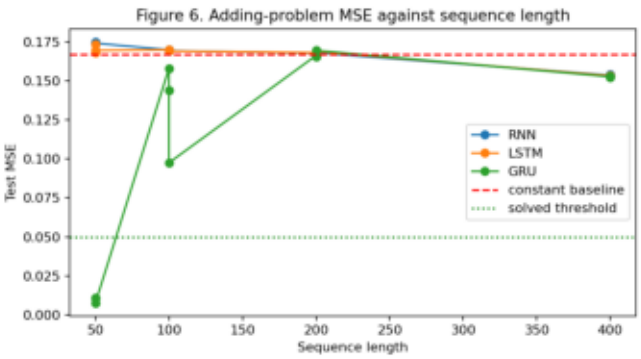
At T=50, GRU achieved 0.00965 +/- 0.00178 and solved 3/3 runs; RNN and LSTM remained near the 0.167 constant baseline. At T=100, GRU mean MSE was 0.13319 +/- 0.03167. At T=200 and 400, all models approached baseline under the fixed 12-epoch budget.

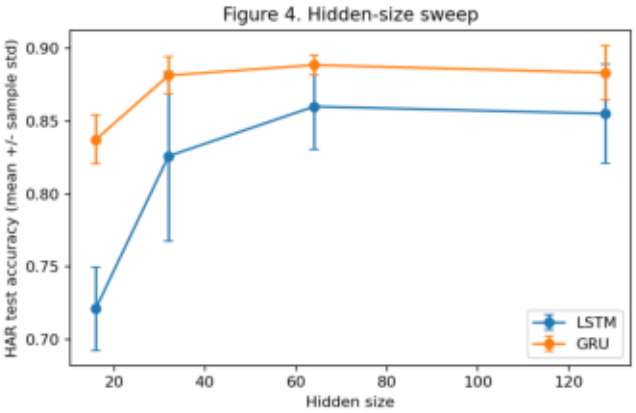
Summary table

T=50: RNN 0.17459, LSTM 0.17044, GRU 0.00965. T=100: RNN 0.16973, LSTM 0.16955, GRU 0.13319. T=200: RNN 0.16812, LSTM 0.16833, GRU 0.16721. T=400: RNN 0.15367, LSTM 0.15337, GRU 0.15264. Each value is mean over three seeds; solved means MSE < 0.05.

Interpretation

The short-sequence GRU learns the marker positions, while long horizons expose recurrent credit-assignment difficulty and vanishing gradients.





Problem 1 - Gradients, capacity and ablation

Gradient study

The T=100 before/after profiles are included. The study is performed in training mode to support cuDNN backward, without an optimizer step. The three models show different gradient attenuation profiles.

Capacity

The hidden-size sweep used 16, 32, 64 and 128 across all three seeds. At h=64, LSTM accuracy was 85.997% with 19,590 parameters; GRU accuracy was 88.847% with 14,790 parameters.

Clipping ablation

No clipping: MSE 0.16812 +/- 0.00025. Norm 1.0: 0.16845 +/- 0.00045. Clipping did not improve this measured setup, so it is reported as an ablation rather than a claimed improvement.

Figure 3. Gradient norm before and after training

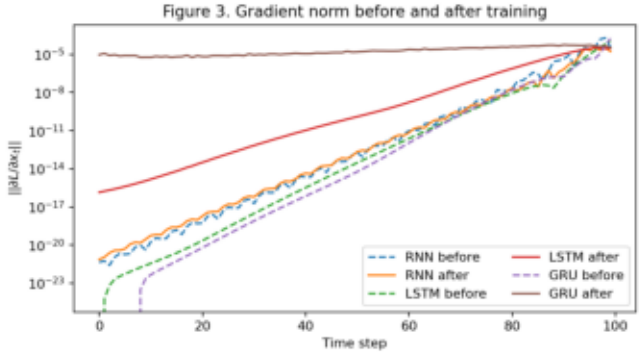
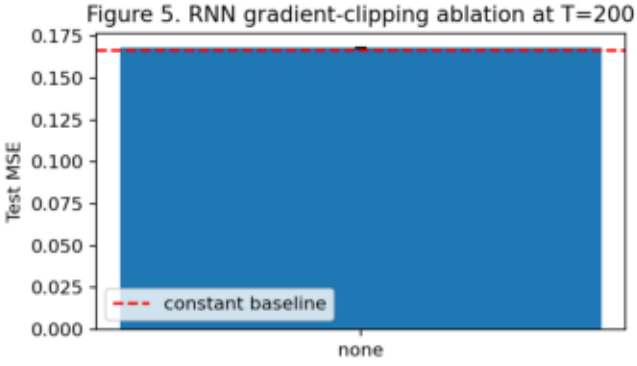


Figure 5. RNN gradient-clipping ablation at T=200



Problem 2 - Attention and Transformer HAR

Hand attention

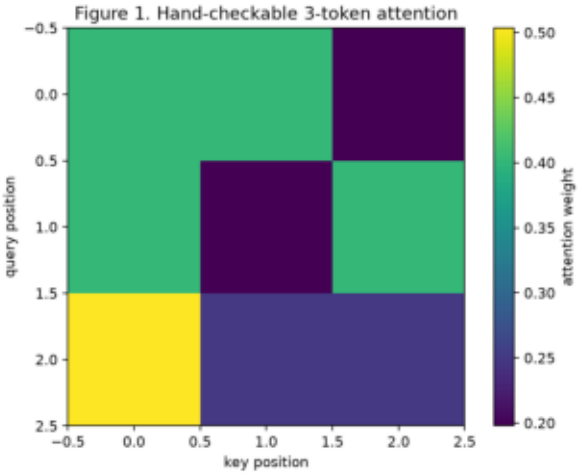
Q=[[1,0],[0,1],[1,1]], K=[[1,1],[1,0],[0,1]], V=[[1,2],[3,4],[5,6]]. The code output is [[2.593327,3.593327],[3.000000,4.000000],[2.489531,3.489531]], matching softmax(QK^T/sqrt(2))V and the framework check.

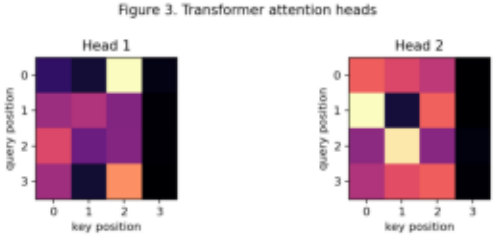
Transformer HAR

None: accuracy 87.659 +/- 1.936%, macro-F1 87.446 +/- 1.980%. Sinusoidal: 89.445 +/- 1.240%, 89.247 +/- 1.303%. Learned: 88.439 +/- 0.859%, 87.984 +/- 0.989%. Each model has 415,494 parameters.

Comparison

Sinusoidal Transformer is competitive with GRU but is much larger. For a constrained wearable deployment, the GRU remains the practical choice.





Problem 2 - Character LM, ablations and conclusion

Language model

The causal model used context 128, width 128, four heads, four layers and 12 epochs. It has 810,049 parameters. Validation loss decreased from 2.3461 to 1.5941 nats, equal to 2.2998 bits per character. Samples at temperatures 0.2, 0.8 and 1.5 are saved in generated_samples.txt.

Ablations

Head counts 1, 4 and 8 were compared at fixed width; parameter count stayed 215,233. Contexts 64, 128, 256 and 512 recorded causal/unmasked timings and peak CUDA memory. The observed memory trend is consistent with O(n^2 d). Unmasked attention is treated only as an ablation, not a valid causal LM.

AI-use and references

AI assistance supported code structuring, debugging, test design, plotting and report organisation. All numerical results are from the author's T4 run. References: Hochreiter & Schmidhuber (1997); Cho et al. (2014); Vaswani et al. (2017); Anguita et al. (2013); Karpathy Tiny Shakespeare; PyTorch documentation.

Figure 2. Tiny Shakespeare validation learning curve

